

A modified method for detecting incipient bifurcations in a dynamical system

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[1] We assess the proximity of a system to a bifurcation point using a degenerate fingerprinting method that estimates the declining decay rate of fluctuations in a time series as an indicator of approaching a critical state. The method is modified by employing Detrended Fluctuation Analysis (DFA) which improves the estimation of short-term decay, especially in climate records which generally possess power-law correlations. When the modified method is applied to GENIE-1 model output that simulates collapse of the Atlantic thermohaline circulation, the bifurcation point is correctly anticipated. In Greenland ice core paleotemperature data, for which the conventional degenerate fingerprinting is not applicable due to the short length of the series, the modified method detects the transition from glacial to interglacial conditions. The technique could in principle be used to anticipate future bifurcations in the climate system, but this will require high-resolution time series of the relevant data. **Citation:** Livina, V. N., and T. M. Lenton (2007), A modified method for detecting incipient bifurcations in a dynamical system, *Geophys. Res. Lett.*, **34**, L03712, doi:10.1029/2006GL028672.

1. Introduction

[2] Bifurcations of the Earth's climate have been the subject of intensive research for several decades, since early works by Lorenz [1963] (for reviews, see, e.g., Ghil *et al.* [2002] and Lockwood [2001]). The highly nonlinear dynamics of the climate are characterised by thresholds and multiple equilibrium states [Dijkstra and Ghil, 2005]. Under the influence of anthropogenic forcing, the climate system may pass through bifurcation points [Alley *et al.*, 2003; Lockwood, 2001], which can indicate collapse of some variable (e.g., thermohaline circulation [Rahmstorf, 1995]) and/or hysteresis behaviour [Stommel, 1961; Manabe and Stouffer, 1988]. Clearly it would be desirable to anticipate such transitions before they occurred.

[3] The analysis of deterministic model equations allows the calculation of explicit analytical solutions and their bifurcations [Abshagen and Timmermann, 2004; Dijkstra and Weijer, 2005]. In models with stochastic components, power spectrum analysis can reveal so-called noise-induced precursors of bifurcation [Wiesenfeld, 1985]. However, with observed climate records, for which a full analytical description is unavailable, the time series methods of general bifurcation analysis are required. While there is open discussion on specific mechanisms leading to climate transi-

tions [see, e.g., Dijkstra *et al.*, 2003], the prognostic study of bifurcations is itself a challenging scientific task. An important case study is the potential collapse of the Atlantic thermohaline circulation (THC). Its weakening has been inferred from observations [Bryden *et al.*, 2005], and models suggest that a bifurcation point exists, but disagree over its proximity [Rahmstorf *et al.*, 2005].

[4] One of the earliest and best-elaborated attempts to employ time series analysis for the detection of climate changes was made by Hasselmann [1997, and references therein] and later by the IDAG consortium [Barnett *et al.*, 2005]. Their optimal fingerprinting method allows one to determine a gradual climate change and to attribute it by means of time series analysis to a particular forcing mechanism. However, to study the proximity of a system to a bifurcation point, more specialized methods are required. The degenerate fingerprinting technique has recently been introduced as a tool for detecting system bifurcations, using a simple autocorrelation function (ACF) to estimate the decay rate in the system [Held and Kleinen, 2004]. Here, we modify the technique to allow analysis of generally correlated climate time series.

[5] Dissipative dynamical systems with extended degrees of freedom can evolve to self-organized critical states with spatial and temporal power-law scaling behaviour [Bak *et al.*, 1988]. Consider a record with short-term temporal correlations, which is characterized by autocorrelation function $C(s) \sim \exp\left(-\frac{s}{s_c}\right)$, where s is lag and s_c is a characteristic length. With increasing s_c ('stretching' of the autocorrelation function), the record can be approximated by a long-term correlated process with autocorrelation function $C(s) \sim s^{-\gamma}$, $0 < \gamma < 1$. Since bifurcations in a time series occur at critical, highly non-stationary states, which are characterized by prominent 'memory' in the data (persistent increase or decrease of the dynamics), a technique estimating the decay rate in the system requires an appropriate modification. A conventional auto-correlation function is a weak tool for such analysis, since it does not provide accurate results in the case of highly non-stationary data.

[6] In recent years, the method of Detrended Fluctuation Analysis (DFA) has become a widely used tool for the study of statistical scaling properties of non-stationary time series [Peng *et al.*, 1994]. It has been applied successfully, e.g., to DNA sequences [Buldyrev *et al.*, 1995], heart-rate [Ashkenazy *et al.*, 2001], and climate dynamics [Kantelhardt *et al.*, 2006]. In essence, the method studies the scaling properties of fluctuations after removing non-stationarities. This is achieved by multiple averaging of the mean-root square variance over windows of variable length in the discrete integral (cumulative sum) of the series. One removes a polynomial trend of predefined order and averages fluctuations over windows of variable size. Data in

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bigger windows have bigger fluctuations around the polynomial trend, producing an increasing DFA curve. Where there are long-term correlations in the data, the DFA curve has a power law exponent which is the slope of linear fit in a log-log plot.

[7] We first introduce the degenerate fingerprinting method used for studying bifurcations in time series, and then describe the DFA-modification of the method. We show how it relates to the initial technique and provide some technical calibration. Then we apply the method to GENIE-1 model output (maximum Atlantic meridional overturning circulation) and to Greenland paleotemperature data in an attempt to detect incipient bifurcations/transitions.

2. Methodology

2.1. Degenerate Fingerprinting

[8] In the degenerate fingerprinting technique, the dynamics of the system are first reduced to a 1D time series, which is then locally modelled by a linear AR(1)-process $y_{n+1} = c \cdot y_n + \sigma \eta_n$, where $c = \exp(-\kappa \Delta t)$ and η_n is the Gaussian white noise. This allows estimation of the decay rate κ which vanishes near the bifurcation point.

[9] The closer the c value to 1 (Figure 1a), the flatter the curve of the autocorrelation function (ACF) and the slower the decay of fluctuations. This property close to the critical state allows one to employ the autocorrelation function to detect an incipient transition in the time series. Moving along the series and estimating the ‘propagator’ c , any trend of c towards the value 1 is an indication of the approach of the bifurcation point.

[10] To calculate the propagator for a given time series, one considers data within a sliding window of fixed length, removes any linear trend from this set, applies Δt -aggregation (summation within non-intersecting windows of length Δt , which eliminates ‘weather noise’ in climatic time series), calculates an auto-correlation function for the aggregated set and estimates the exponent of the decay. Δt is chosen such that $\Delta t \gg 1/\kappa_i$, where κ_i are decay rates of minor bifurcation modes, which are insignificant for our analysis (thus removing the ‘weather noise’), and $1/\kappa \gg \Delta t$ where κ is the decay rate of the major mode [Held and Kleinen, 2004]. In our case, $\Delta t = 50$.

[11] The evolution of the ACF-propagator, while the window slides along the series, shows the dynamics of the system. A window length equal to 10% of the length of the series is an appropriate choice: bigger windows would provide poor final statistics of the propagator, whereas smaller windows would be insufficient to estimate the decay rate using the ACF. This method was successfully applied to CLIMBER-2 model salinity fields of time length 50,000 years [Held and Kleinen, 2004].

2.2. Detrended Fluctuation Analysis (DFA)

[12] Real geophysical systems carry memory (e.g., air temperature, SST, river flux), caused by different types of inertia or trends. Physically, this temporal scaling means that the state of the system depends not only on the previous time step, but on the history with some time lag. Statistically, it is described in terms of correlations, and one of the first studies of correlations in climate can be found in [Koscielny-Bunde et al., 1998]. A time series that possesses long-range temporal correlations can be characterized by

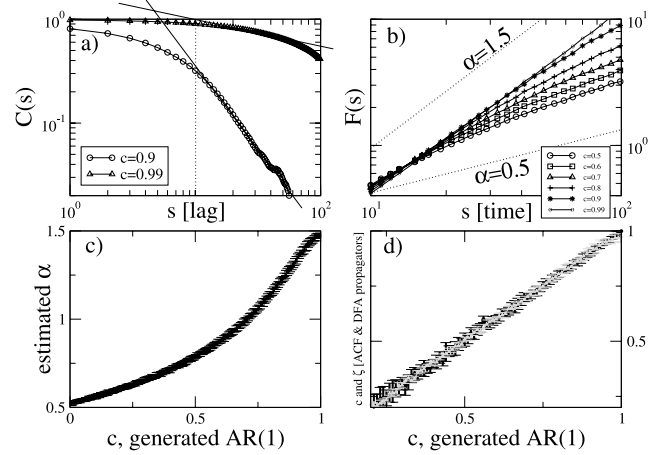


Figure 1. (a) Auto-correlation function for AR(1) data, $c = 0.9, 0.99$; as $c \rightarrow 1$, the exponential decay becomes comparable to a power-law. (b) DFA curves for AR(1) data; as $c \rightarrow 1$, the DFA exponent $\alpha \rightarrow 1.5$. (c) DFA exponents for AR(1) data, $0 \leq c < 1$ with step 0.001. For each c , twenty samples of length 20K generated to obtain error bars. (d) DFA1-propagator ζ and ACF-propagator c for AR(1) data of length 20K, twenty samples for each c provide error bars; gray (ACF data) and black (DFA data) curves are almost identical, indicating the accuracy of the calibration.

scaling laws with a relevant scaling exponent, and there exist several methods for estimating the correlations. We will operate with basic power-spectrum exponent β , auto-correlation exponent γ , and DFA exponent α , which are all connected by analytical relations given below.

[13] The power spectrum $S(f)$, where f is a frequency, in a long-range correlated time series has a power law form with a scaling exponent β (i.e., the power spectrum is proportional to $1/f^\beta$). Similarly, the auto-correlation function $C(s)$, where s is a time lag, provides power-law exponent γ . Trends that exist in climate records may cause inaccuracies in measuring these two exponents. Hence more advanced techniques that exclude trends from the data have been developed. One of the methods that copes well with non-stationarity is the Detrended Fluctuation Analysis (DFA).

[14] In this method, we integrate time series x_i and divide the range of the obtained series (the profile, i.e., discrete integral/cumulative sum of x_i) into windows of size s . Next, within each window, we calculate the best polynomial fit of chosen order and evaluate the square of the difference between the polynomial and profile function. At the next step, we average the obtained values over all windows and repeat the procedure for different window scales s , arriving at ‘variance’.

$$F^2(\nu, s) = \frac{1}{s} \sum_{i=1}^s [Y((\nu - 1)s + i) - y_\nu(i)]^2, \quad (1)$$

where Y is the profile function of initial time series x_i , $y_\nu(i)$ is the best fitting polynomial of order k in segment ν , while ν is the index of non-overlapping windows of size s , and N is the length of the time series. Then we average over all

segments ν and take the square root to obtain the fluctuation function $F(s)$.

[15] If the series is long-term power-law correlated, the DFA fluctuation function increases by a power-law:

$$F(s) \propto s^\alpha, \quad (2)$$

where α is the DFA scaling exponent. For uncorrelated records, $\alpha = 0.5$, while for long-term correlated (persistent) records, $\alpha > 0.5$ [Kantelhardt *et al.*, 2001]. In DFA of k th order, trends of polynomial order $k - 1$ are eliminated. The DFA exponent α is related to the power spectrum exponent β from $S(f) \sim 1/f^\beta$ by $\beta = 2\alpha - 1$, such that for white noise with $\alpha = 0.5$, the power spectrum is flat with $\beta = 0$, and for a random walk signal (integrated white noise) with exponent $\alpha = 1.5$, the power spectrum indicates red noise with $\beta = 2$. The auto-correlation exponent γ in $C(s) \propto s^{-\gamma}$ (here s is the ACF lag) is related to the DFA exponent α by $\alpha = 1 - \gamma/2$, where $0 < \gamma < 1$ (stationary series).

2.3. Modified Method: DFA-Propagator

[16] At critical states close to the bifurcation, when the propagator $c \rightarrow 1$, the slow exponential decay is well approximated (in terms of absolute deviations in the log-log plot of $C(s)$) by a power law in the short-term regime (10–100 time units), with auto-correlation function exponent $\gamma \rightarrow 0$ (see Figure 1a, where γ is the slope of the fitted lines). (In the long-term regime, an AR(1) process is not power-law correlated, nor is the exponential decay equivalent to the power-law decay.) Hence in the short-term regime, detrended fluctuation analysis can be used for estimation of the propagator (we consider the same time scale for DFA). The DFA exponent α is estimated as a coefficient of linear regression in a log-log plot of $F(s)$ vs s . Figure 1b shows that the AR(1)-process becomes non-stationary at critical values of c , and when $c \rightarrow 1$, the DFA exponent $\alpha \rightarrow 1.5$, which corresponds to a random walk state of the system. Note, however, that this kind of non-stationarity is a necessary but not sufficient condition for bifurcation. In the case of a ‘blind’ test (when we do not know whether the series collapses or not), the propagator trend should be considered as a warning of potential system bifurcation.

[17] The advantage of DFA is that it avoids artifacts (short-term ‘weather noise’), because the DFA exponent is measured in time scale above 10 time units. In contrast, the conventional ACF lag-1 correlations are related to the shortest time scale and are more vulnerable to the influence of noise, and therefore the aggregation of the series is required. Therefore, the series should be long enough to allow aggregation, and short records cannot be studied with the ACF-based technique.

[18] When compared to the auto-correlation power-law exponent γ , the DFA does not have boundary restrictions (imposed on γ by the stationarity condition) and it therefore provides more flexibility in the estimation of correlations. The series should have at least several hundred points per window, and a sufficient number of sliding windows to properly estimate the evolution of the propagator. Therefore, the record should be at least of order a thousand points. To detect an incipient bifurcation, one could monitor the dynamics of the DFA exponent α towards its critical

value 1.5 which is approximately equivalent to critical c value 1 (non-stationary state, which leads to bifurcation).

2.4. Calibration of the DFA-Propagator

[19] Here, to make the DFA exponent α comparable with the ACF-propagator, we take the additional step of calibrating it towards c using polynomial regression and call it the DFA-propagator, with notation ζ . (Note that when studying a series with unknown dynamics and high nonlinearities, the present calibration might be inappropriate.) We have simulated AR(1) data of length 20,000 and found a relationship between the exponents. Initial value $y(0)$ was taken randomly from a Gaussian sequence with zero mean and standard deviation $1/\sqrt{1 - c^2}$.

[20] The suggested empirical calibration is as follows: $\alpha = 0.91c^3 - 0.37c^2 + 0.49c + 0.52$ for $0 < c \leq 0.936$; $\alpha = -12.38c^2 + 25.14c - 11.28$ for $0.936 < c \leq 0.967$; $\alpha = 0.72c + 0.75$ for $0.967 < c < 1$. Exponent α is first estimated in the data and then transformed using the above calibration to obtain the DFA-propagator ζ . In the case of highly correlated data ($\alpha > 1.5$), the values of the DFA-propagator ζ may exceed 1, because the DFA exponent is not restricted by the stationarity condition valid for ACF. Once the propagator has reached the critical value 1, the higher correlations just speed up the process of transition.

2.5. Testing the Propagators on AR(1) Data

[21] After the calibration, which provides comparable values of ACF- and DFA-propagators, we simulated AR(1) data of length 20,000 with ensembles of twenty samples for each value of exponent. In Figure 1d, we show calibrated DFA-propagator ζ against similarly calculated ACF-propagator c , and error bars for the DFA-propagator ζ are shown without transformation, as they were for the DFA exponent α . Both methods are comparable in accuracy. Similar analysis for much shorter data (200 points) reveals that the ACF-propagator deviates more at lower c values, but has smaller error bars close to the critical value $c = 1$, which is an advantage of the conventional technique. In the case of less well-behaved real data (possibly more highly correlated), the techniques need to be tested, compared again, and the DFA-propagator ζ re-calibrated if necessary.

3. Data and Results

[22] The first data we used for analysis were generated from the GENIE-1 model, using the GOLDSTEIN ocean [Edwards and Shepherd, 2002] coupled to an Energy-Moisture Balance atmosphere model (EMBM) and sea-ice, i.e. the C-GOLDSTEIN configuration [Edwards and Marsh, 2005]. The ‘traceable’ parameter set of [Lenton *et al.*, 2006] was used. The model was forced with a gradual increase of atmospheric CO_2 and an additional stochastic freshwater flux perturbation. The latter was introduced because the simple atmosphere alone produces little or no variability. The CO_2 concentration was increased gradually from the preindustrial level of 278 ppm at a rate of 0.003% per model year, producing about 1250 ppm after 50,000 years. This forces a collapse of the thermohaline circulation after about 38,000 years. As data, we consider the yearly maximum Atlantic meridional overturning circulation stream function. In Figure 2a, we show five model samples with different

amplitude of stochastic freshwater flux perturbation. We applied the modified degenerate fingerprinting method to these model runs (Figure 2b) and plotted the values mapped in abscissa to the end points of the windows to make the analysis similar to real-time data monitoring. The DFA-propagator shows a trend towards the critical value at 35,000–40,000 years with a characteristic kink near the bifurcation point due to abrupt change of the data pattern at 38,000 (denoted by the vertical dotted line). This reflects discontinuity in the data which causes instant loss of memory at the point of bifurcation and is detected by the DFA-propagator. Such a kink might be an additional indicator of collapse.

[23] We have also considered Greenland ice core paleotemperature data available on the internet (GISP2 project [Alley, 2000]). This spans the time from 50 kyr ago to present, but the series comprises only 1586 points. Here, the DFA-propagator ζ was calculated in sliding windows of length 500, and since the data were unevenly spaced, this was, besides the poor statistics, an additional complication of the analysis. However, even in such conditions, the DFA-propagator is able to indicate the transition in the series (Figure 3): the warming at the end of the Younger Dryas around 12 kyr before present (calendar calibrated years) is anticipated by an upward trend in the propagator, which reaches 1 around this time. Note that the ACF-propagator c is not applicable for such a short record, which cannot be aggregated. Strictly speaking, the temperature shift from colder (-45°C) to warmer (-30°C) state in the Greenland record might not be a bifurcation in the sense of a phase diagram, but locally, the rapid temperature increase is a critical behaviour, and is well detected (Figure 3). D-O

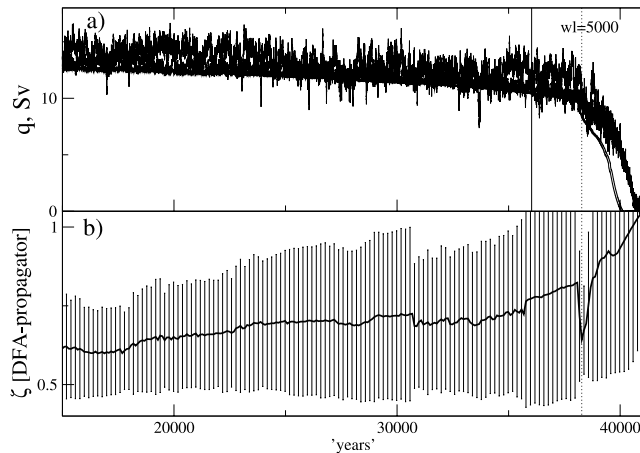


Figure 2. (a) Five runs of GENIE-1 maximum overturning stream function, different magnitude of stochastic freshwater flux perturbation. (b) DFA-propagator ζ with error bars for the runs in Figure 2a calculated in sliding windows of length 5,000 and mapped into the end points of the windows. The dotted vertical line denotes the point of the data pattern change which is detected as the kink in the propagator. The solid vertical line in Figure 2a denotes the border of the last sliding window of length 5,000. We apply DFA1 with minimal detrending, because trend might contribute into the transition dynamics, and eliminating it might postpone detection of the bifurcation.

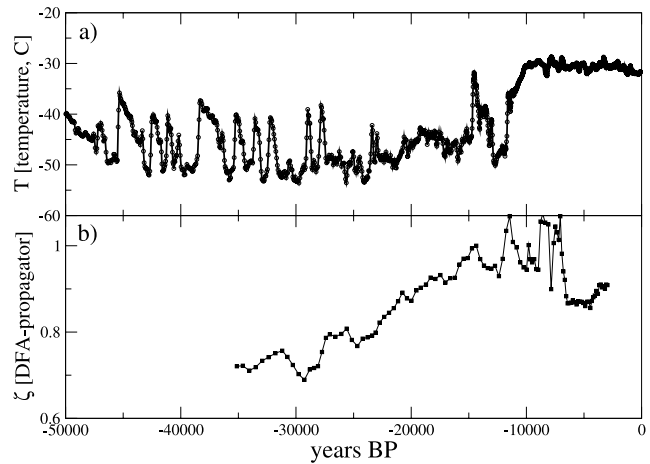


Figure 3. (a) GISP2 Greenland paleotemperature (unevenly spaced record, which is seen in different density of symbols on the curve). (b) DFA1-propagator ζ calculated in sliding windows of length 500 points (non-equidistant time) and mapped into the middle points of the windows.

events that are present in the early part of the time-series are insufficiently resolved to be detected.

4. Discussion

[24] From our study of model and paleo data we suggest that the modified degenerate fingerprinting technique is suitable for bifurcation analysis of climatic time series, and could in principle be used to forecast potential climate system bifurcations. However, the following problems arise: Since each propagator value is based on an average within a window, attribution of the bifurcation to a particular time moment is not certain. Moreover, for more reliable analysis, it is preferable to consider longer time windows, and this, accordingly, increases the uncertainty in timing the bifurcation. A further problem is that in real-time climate monitoring, one can only be certain about the occurrence of a transition at (or very close to) the critical point. The earlier trend of the propagator only provides a hypothetical prediction of the transition, which might be avoided by feedbacks or changes in forcing.

[25] As both versions of the degenerate fingerprinting technique are based on a linearised approximation, in cases of complicated nonlinearities, predictions of bifurcations may fail. However, the method at least provides an initial basis for bifurcation prognosis in the climate system. The technique should prove useful for the bifurcation analysis of model forecasts of future climate evolution. Ideally it would also be applied to real data in an attempt to anticipate any future bifurcations in the climate system. This requires a time series that is an order of magnitude longer than the transition time of the system of interest, with an order of magnitude higher time resolution. Such records are generally lacking at present, for example, despite some searching we have failed to find a diagnostic time series for the Atlantic meridional overturning circulation (e.g., sea surface temperature) that has the requisite combination of duration and resolution. However, we anticipate that increasing amounts of data will become available to test the predictive abilities of the method.

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